

Callie Yejin Kim

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EDUCATION

PhD, Computer Science

Present

University of Wisconsin-Madison, 3.55/4.00 cumulative GPA

Madison, WI

Advisor: Dr. Bilge Mutlu

M.S, Computer Science

University of Maryland, 3.92/4.00 cumulative GPA

College Park, MD

Advisor: Dr. Huaishu Peng

B.S, Computer Science and Engineering

Ewha Womans University, 3.78/4.00 cumulative GPA

Seoul, South Korea

PUBLICATIONS

* indicates equal contribution

Kim, C., White, N., He, E., Sala, F., & Mutlu, B. RoboCritics: Enabling Reliable End-to-End LLM Robot Programming through Expert-Informed Critics, In *ACM/IEEE Human Robot Interaction (HRI 26)*. **Honorable Mention Award** 23.2% Acceptance Rate

Johnson, H. L., Spalitta, H., **Kim, C.**, & Mutlu, B. Supporting Money Management among Adults with Down Syndrome: A Multi-Technology Probe Study, In *ACM Human Factors in Computing Systems (CHI 26)*.

Koike, A., Guo, S. G., He, X., **Kim, C.**, Sullivan, D., & Mutlu, B. Elements of Robot Morphology: Supporting Designers in Robot Form Exploration, In *ACM/IEEE Human Robot Interaction (HRI 26)*. 23.2% Acceptance Rate

Kim, C., Sato, A., White, N., Ho, H., Lee, C., Hwang, Y., & Mutlu, B. Bridging Generations using AI-Supported Co-Creative Activities, In *ACM Human Factors in Computing Systems (CHI 25)*. **Honorable Mention Award** 24.9% Acceptance Rate

Kim, C.*, Lee, C.*, & Mutlu, B. Understanding Large-Language Model (LLM)-powered Human-Robot Interaction, In *ACM/IEEE Human Robot Interaction (HRI 24)*. 24.7% Acceptance Rate

RESEARCH EXPERIENCES

Graduate Research Assistant - People and Robots Laboratory

Madison, WI

Advisor: Dr. Bilge Mutlu

Sep 2021 - Present

- Conducted research on end-user robot programming and task specification, designing systems that enable non-expert users to effectively specify, verify, and refine robot behaviors.

- Developed a robotics framework combining LLM-based task specification with expert-informed motion-level critics to improve the reliability of robot behaviors.
- Evaluated a block-based robot programming system in community workshops, studying paired end-user workflows.

University of Maryland - Dept. of Computer Science

College Park, MD

Advisor: Dr. Huaishu Peng

July 2020 - May 2021

- Designed and fabricated wearable VR haptic prototypes using CAD-designed silicone components integrated into custom masks to support accessible spatial understanding.

TEACHING EXPERIENCE

Lecturer, University of Wisconsin-Madison, CS502: Theory & Practice of CS Education (Sept 2025)

Teaching Assistant, University of Wisconsin-Madison, CS400: Programming III (Jan 2022 - Spring 2025); CS537: Intro to Operating Systems (Aug 2021 - Dec 2021)

Teaching Assistant, University of Maryland, CMSC425: Game Programming (Aug 2020 - May 2021)

INVITED TALKS

October 12th, 2024, Large-Language Models (LLM) for Human-Robot Interaction. Mentorship Program on HRI and Robot Learning, *University of Virginia*

PATENT

Sangsoo Park, Callie Y. Kim, Ina Shin, and Hyunkyung Jung. Virtual Reality Based Disaster Education Method, Device and Computer Readable Medium for Performing the Method. KR Patent Application No. 1020180160585, Registration No. 1021139260000

SERVICES

Conference Review: CHI, UIST, DIS, HRI, ICRA, IEEE VR

Journal Review: IEEE Transactions on Human-Machine Systems

Grandparents University, Instructor, University of Wisconsin-Madison

Annually: July 2022, 2023, 2024

SKILLS

Research Methods: System Prototyping, Participatory Design, Semi-Structured Interviews, Community workshops, Thematic Analysis, Surveys, Mixed-Methods Evaluation

Programming Languages: Python, Java, C#, Javascript, HTML, CSS, Swift

Tools and Frameworks: React, ROS, Gazebo, RViz, Unity, Flask, Docker, PyTorch, LangChain

AI Techniques and Computational Methods: Large Language Models, Computer Vision (Object Detection), AR/VR